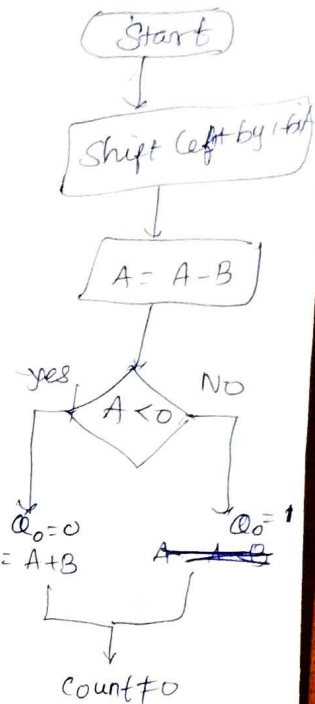


Divident (D) $\Rightarrow 9 \Rightarrow 1001$ ~~restoration~~ (restoring division)

Divisor (B) $\Rightarrow 3 \Rightarrow 00011 \Rightarrow 2's(3) \Rightarrow 1's \Rightarrow 11100$
 $2's \Rightarrow 11101$

A $\Rightarrow$ Size(B) $\Rightarrow 00000$

Pass	A	Q	operation
1	00000 $\begin{array}{r} \boxed{\times} 00001 \\ 11101 \\ \hline A=A-B \quad 11110 \\ \downarrow A < 0 \\ Q_0 = 0 \\ A = A+B \\ \begin{array}{r} 11110 \\ 00011 \\ \hline \boxed{\times} 00001 \end{array} \end{array}$	1001 + 0 1001 $001 \boxed{0}$ 0010	* Shift left by 1-bit * $A = A - B$ * $A < 0$ * $Q_0 = 0$ * $A = A + B$



2	00001 Shift left $\begin{array}{r} \boxed{\times} 00010 \\ 11101 \\ \hline A=A-B \quad 11111 \\ \downarrow A < 0 \\ Q_0 = 0 \\ A = A+B \\ \begin{array}{r} 11111 \\ 00011 \\ \hline \boxed{\times} 00010 \end{array} \end{array}$	0010 $010 \boxed{0}$ 0100	Shift left by 1-bit $A = A - B$
3	00010 Shift left $\begin{array}{r} \boxed{\times} 00100 \\ 11101 \\ \hline \boxed{\times} 00001 \\ \downarrow A > 0 \\ Q_0 = 1 \end{array}$	0100 0100 $100 \boxed{1}$	Shift left by 1-bit $A = A - B$

~~No~~ restoration

00001

1001

4

00001
Shift left by 1-bit

~~0~~ 00011
11101

X 00000

$A > 0$

$Q_0 = 1$

00000

Remainder

1001

0011

0011

Quotient

Shift left by 1-bit

$A = A - B$